CGA: Conditioning as Group Action

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Abstract

Condition a network by rotating its features, with rotation angles linear in the condition, and combined conditions compose automatically: rotations add, so the network handles combinations it never saw in training. FiLM and hypernetwork conditioners fail on the same combinations at up to 43× the cost. We present **CGA** (Conditioning as Group Action), a framework that treats feature conditioning $y = T(c)\,x+\beta(c)$ as a group representation of the condition space: the condition c enters the operator linearly in an abelian Lie algebra, $T(c) = P \exp(\mathrm{A}(Wc))P^{\top}$, so the homomorphism law $T(c_1+c_2) = T(c_1)T(c_2)$ holds exactly, for any weights, by construction. Choosing the algebra recovers deployed mechanisms as special cases. The diagonal algebra yields an exactly compositional variant of FiLM; FiLM's usual MLP head parametrizes the same orbit without the homomorphism, and our ablations show this missing law is what fails under composition. The maximal torus $\mathfrak{so}(2)^{d/2}$ with scalar $c{=}n$ recovers RoPE, and with learned generators the multiplicative GRAPE construction, transplanted from attention positions to arbitrary condition vectors in FiLM's role. Our instantiation, planar rotations with angles Wc conjugated by a Group-and-Shuffle structured orthogonal basis, is exactly orthogonal, equals the identity at $c{=}0$ and at initialization, and costs 1.11× FiLM's conditioning FLOPs; a dense learned basis provably exceeds any FiLM-comparable budget, and transformer sites need no explicit basis at all, as in RoPE. Under a pre-registered, budget-fair protocol against FiLM, conditional LayerNorm, concatenation, and two input-conditioned hypernetworks (up to 48× our parameters), CGA achieves the lowest compositional OOD error on all five transformation suites, with complete seed separation throughout ($p{=}9.1{\times}10^{-5}$, $N{=}10$), including zero-shot extrapolation to compositions longer than any seen in training. Swapping only the linear map for an MLP degrades OOD error by up to $4.3{\times}10^4$. The most expressive baselines generalize worst. Three pre-registered gates returned negatives that bound the method: a 1.46× in-distribution fit penalty on rich scenes; a content-conditioning failure in diffusion (a rotation preserves information, so it transforms representations and cannot inject content); and no advantage in recurrent world-model rollouts, where representation-consistency error dominates operator-composition error and accumulates under isometries that a contractive transition would absorb. We report all three as registered. Exact composition is a guarantee about the operator, and the negatives locate where operator guarantees stop paying.

1 The framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM [1] computes $y = \gamma(c) \odot x + \beta(c)$; adaLN in DiT blocks [2] is its normalized form; hypernetworks emit $W(c)$ [3]. All are instances of

$$y = T(c)\,x + \beta(c), \tag{1}$$

and they differ only in the family from which the per-sample operator $T(c)$ is drawn.

Definition 1 (Compositional conditioner)**.** *$T : \mathbb{R}^k \to \mathrm{GL}(d)$ is* compositional *if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1{+}c_2) = T(c_1)\,T(c_2)$ and $T(0) = I$.*

Proposition 1 (Exact composition by construction)**.** *Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $\mathrm{A} : \mathbb{R}^m \to \mathfrak{a}$ linear, $W \in \mathbb{R}^{m\times k}$, and P orthogonal. Then $T(c) = P \exp(\mathrm{A}(Wc))\,P^{\top}$ is a compositional conditioner for every W, P; if $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.*

Proof. $\exp(X)\exp(Y) = \exp(X{+}Y)$ for commuting X, Y; $\mathrm{A} \circ W$ is linear; conjugation preserves products and $\exp(0) = I$. □

Composition is a property of the parametrization: it holds at initialization, during training, and off-distribution. The learnable parts, W and P, select which one-parameter subgroups the condition drives and how fast.

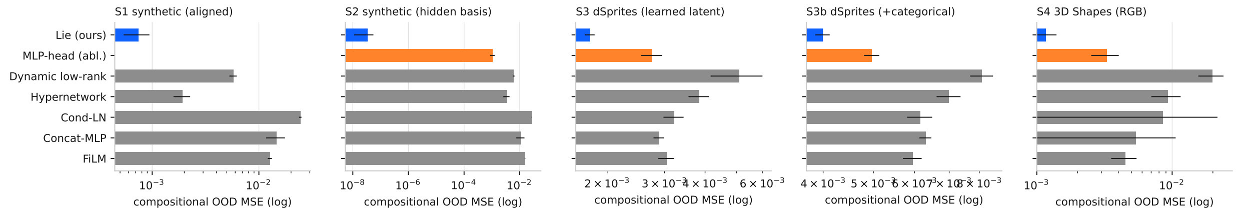

Figure 1: OOD error by conditioner (log scale; blue = CGA, orange = MLP-head ablation).

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms exactly:

$\mathfrak{a}$	condition	P	recovered mechanism
$\mathrm{diag}(\mathbb{R}^d)$	c	I	*exponential FiLM*: $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	**this work** (FiLM's role)

Standard FiLM parametrizes the same diagonal orbit through an MLP head $\gamma(c)$: the orbit matches, the homomorphism is absent, and §2 shows this missing law is what fails under composition. Our ablation with an MLP angle head (same orbit, same basis, more compute) loses by up to 4.3×10^4. The PEFT literature (LoRA, OFT, BOFT, HRA, Group-and-Shuffle [6–10]) owns these operator families as fixed weight adaptations; CGA places them in FiLM's role, per-sample and in activation space, where the composition question exists.

Cost, and where the basis comes from. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs and no matrix exponential. The basis carries the entire budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is 1.52× FiLM's whole conditioning path at d=128 and exceeds any FiLM-comparable ceiling. Two resolutions work. **(i)** A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to 1.11× FiLM; P is undercomplete (4,800 parameters vs. $\dim \mathrm{SO}(128)$=8,128) yet suffices, because P need only expose the planes the condition acts on. **(ii)** Transformer sites need no explicit P: the adjacent dense projections absorb the basis, which is how RoPE deploys. The operator satisfies identity initialization (W=0 $\Rightarrow T \equiv I$), $T(0) = I$ structurally, and $\sigma(T(c)) \equiv 1$; unit tests pin all three invariants in the release.

2 Experiments

Protocol. Every suite is pre-registered: margins, tests (N=10 seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff's $|\delta| \geq 0.474$), and single-read OOD splits are fixed before each run. All arms share identical backbones. A shared counter enforces conditioning cost ≤ 1.20× FiLM and parameters ≤ 1.05× the smallest hypernetwork baseline; baselines may be larger, and reach 48×. We report every verdict as registered, including the negatives. Baselines: FiLM, conditional LayerNorm, Concat-MLP, dense hypernetwork $I+\Delta W(c)$, dynamic low-rank $I+A(c)B(c)^\top$, plus an MLP-head ablation (identical basis, more compute, no linearity).

Suites. S1/S2 (synthetic): learn $y = M(c)x$ for multi-hot compositions of 8 rotation primitives; S2 hides the primitives' planes behind a random basis B and additionally tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** conditional image transformation. The model encodes a real image, applies $T(\Delta)$ on the learned latent for a factor-change Δ, and decodes; pixel MSE against the dataset's unique ground-truth image scores the result. OOD means never-trained two-factor change types; S3b/S4 include a categorical (shape) factor encoded as a one-hot difference. **S5 (content role):** a width-128 DiT trained as a factor-conditional diffusion model on dSprites; full conditioning determines the target image uniquely, so held-out-combination generation is scored by pixel MSE against ground truth. **S6 (rollout role):** an action-conditioned world model in which the conditioning arm is the latent transition, applied recurrently; OOD means unseen action-type pairs and rollout horizons 10 and 20 with training capped at 3.

Table 1: Compositional OOD error (mean over 10 seeds). CGA is lowest in every column with complete seed separation vs. the strongest hypernetwork baseline (p=9.1×10^{-5}, δ=−1.0). S1–S3b pass the full pre-registered gate; S4 returns a registered negative on a separate in-distribution criterion (text).

Conditioner	S1	S2	S3	S3b	S4
FiLM	1.3×10^{-2}	1.6×10^{-2}	3.0×10^{-3}	5.9×10^{-3}	4.5×10^{-3}
Concat-MLP	1.5×10^{-2}	1.1×10^{-2}	2.9×10^{-3}	6.3×10^{-3}	5.4×10^{-3}
Cond-LN	2.4×10^{-2}	2.7×10^{-2}	3.2×10^{-3}	6.1×10^{-3}	8.6×10^{-3}
Hypernetwork	1.9×10^{-3}	3.4×10^{-3}	3.8×10^{-3}	7.0×10^{-3}	9.3×10^{-3}
Dyn. low-rank	5.7×10^{-3}	6.1×10^{-3}	5.1×10^{-3}	8.1×10^{-3}	2.0×10^{-2}
MLP-head (abl.)	–	1.1×10^{-3}	2.7×10^{-3}	5.0×10^{-3}	3.3×10^{-3}
Lie (ours)	7.4×10^{-4}	3.3×10^{-8}	1.8×10^{-3}	4.0×10^{-3}	1.2×10^{-3}

Findings. **(1) Group-structured conditions: solved.** On S2, CGA reaches 3.3×10^{-8} OOD MSE, five orders below the best baseline, and 5.3×10^{-8} on never-trained triples: error stays flat as composition length grows. Learned angles recover the true generators to 2×10^{-4} rad. A coordinate-aligned ablation (P=I) collapses to FiLM's error, so basis discovery carries the result. **(2) The advantage survives learned latents.** On dSprites every arm fits in-distribution identically, which isolates composition: CGA's OOD/in-dist gap is 1.25× (1.63× with the categorical factor) vs. 2.1–3.3× for every baseline, a 53.8% (42.9%) margin over the strongest baseline at 39× fewer conditioning FLOPs. **(3) Linearity is the mechanism.** The MLP-head ablation shares the basis and the orbit and spends more compute; it loses by 4.3×10^{4}× on synthetic triples and 1.55× on dSprites. **(4) Capacity inverts.** The two hypernetwork arms can represent any operator, and they post the worst compositional gaps on images (2.7–26×): the extra expressivity memorizes training condition types. **(5) The registered negatives.** On 3D Shapes, CGA posts the best OOD error (87.5% margin; gap 1.13× vs. 13–26×) and pays a 1.46× in-distribution fit penalty that trips the pre-registered no-regression gate ($\leq$ 1.10×). In the content suite (S5), rotation conditioning underfits by 7× when the condition must specify image content, and the MLP-head ablation fails identically (0.31 vs. film's 0.04): the failure sits in the rotation channel itself, which preserves information and therefore transforms representations well and injects content poorly. In the rollout suite (S6), every arm compounds error identically over never-trained horizons (all reach ≈0.043 at length 20), and the mildly contractive dynamic low-rank arm degrades least: rollout error is dominated by representation-consistency error, which contraction absorbs at each step and which an isometry preserves indefinitely; exact operator composition does not repair an inconsistent latent. We report all three suites as negatives, as registered. The protocol prices the guarantees rather than assuming they pay everywhere.

3 Discussion

CGA does for conditioning what RoPE and GRAPE did for position: it moves composition from a behavior the network must learn into a law the parametrization guarantees. The evidence here is deliberately controlled (64×64 images, known factors); the deployment experiment is the adaLN site of diffusion transformers, where the basis-free variant drops in below FiLM's marginal cost. Four boundaries are already quantified: categorical factors attenuate the advantage; richer scenes price the orthogonality constraint in in-distribution fit; content specification defeats a transformation channel; and recurrent rollouts are limited by representation consistency, which no operator guarantee repairs. The first three motivate the abelian-algebra generalizations of Proposition 1 beyond the pure torus (torus × diagonal: complex-valued FiLM, with magnitude as the content channel and phase as the composition channel) as the next design point; the fourth motivates pairing the operator with a representation consistency objective. Adjacent work learns group actions on latents for world models [11, 12]; CGA lives in FiLM's role under budget parity, and its composition law holds by construction instead of by training objective.

Reproducibility. Every suite is pre-registered; every number regenerates from append-only logs; unit tests pin the operator invariants and the fairness counters; the repository discloses all amendments and one accounting erratum.

References

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